

backend/api/distillation.py Documentation (Roman Urdu)

Yeh file application ke subse powerful hissey "Context Distillation" (Bara data chota ar asan alfaaz me convert krna) ka bridge/API route ha.

File Ka Maqsad (Purpose)

Isma wo sab links hain jo Frontend setting panel aur graph analyzer istamal krty hta k AI ki thinking quality (compression ratio, verification, confidence) ka record dekh skkein.

Code Blocks aur Endpoints ki Tafseel

1. GET / PUT /settings

- **Kya karta hai?:** User web interface se Distillation, Chain-of-Density ya Compression ko ON/OFF krey to ye endpoints configuration file ko database me update karty hain.

2. POST /query aur /analyze-query

Kya karta hai?: Ye normal chat nahi hai bulkay Advanced RAG ka hissa hy. Jab user koi file upload kry ar usmy se deep search kry toh idhar request ati h aur backend ye calculations krta ha:

- *Compression Ratio* kitna rha? (Misal: 1000 words ko 200 ma krdia).
- *Sub-Queries*: "Python features aur syntax btao" ko d0 sawalon ma todna.

3. POST /chain-of-density aur /verify

Kya karta hai?:

- **Density:** Yeh Ollama ko content bhej kr 5 dafa (5 iterations) mzid density / summary nikalwata ha ta k gaarha point bachay.
- **Verification:** AI apna e jawaab cross-check karra ha k usnyn references s thek prha ha ya ghalatian (hallucinations) ni hn.

4. GET /metrics

- **Kya karta hai?:** Frontend ky pie-charts aur graphs ke liye data fetch krna, ky ajtak system ne kitni bar acha compression kia or average score kitna hy.